

Incorporating NRL Team Lineups into Match Prediction Systems (2012–Present)

Executive summary

Lineup information can explain a large share of match-to-match variance that match-history features cannot capture, especially when high-impact spine players appear or withdraw late. The key technical shift is to treat “team strength” as a sum of (i) a team baseline that evolves over time and (ii) a lineup component that depends on *who* plays and *where* they play (starting 13 vs interchange vs reserves), plus uncertainty because the named squad is not the final 17. ¹

NRL lineup data exists in public, primary sources, but access is shaped by platform constraints and licensing. Official club and league sites publish squad lists on a schedule that has changed over time: in 2012 the league described Tuesday lists updated with “Friday morning late mail”, and later communications in 2017 and 2019 describe 21-man Tuesday squads updated to 19-man 24 hours pre-game and 17-man one hour pre-kick-off; a 2026 club post shows a 22-man Tuesday squad at 4pm. ²

The most robust modelling pattern for your setting (games 2012–present; existing match-history features) is:

- (1) build a canonical historical lineup table (match_id × player_id × team_id × listed_position × status),
- (2) estimate player and position effects with strong regularisation or Bayesian shrinkage to avoid overfitting,
- (3) integrate “Thursday inference” by marginalising over uncertain final selections using historical selection probabilities from named squads, and
- (4) validate with time-respecting splits, calibration, and ablations that isolate what lineups add beyond team form and head-to-head history. ³

Reinforcement learning (RL) is useful once you can (a) simulate counterfactual lineups or (b) frame selection and interchange decisions as policies with measurable rewards. Given rugby league data scarcity, offline RL is the relevant regime: it needs careful off-policy evaluation (OPE) and conservative policy improvement to avoid learning actions unsupported by historical data. ⁴

Lineups in the NRL data lifecycle

Squad announcements are not a single event. The data product you need depends on whether you predict on Tuesday, Thursday, or just before kick-off. In 2012, league posts describe weekly team lists “announced first every Tuesday” and “updated with the Friday morning late mail”. ⁵ A club explainer from 2017 describes a three-stage process: 21 players named Tuesday (4pm AEST), reduced to a 19-man squad 24 hours pre-game, then final 17 confirmed one hour pre-kick-off, with limited exceptions approved by the NRL Head of Football. ⁶ A league post from 2019 repeats the 21-man → 19-man → 17-man cadence explicitly for that round. ⁷

The 2026 season shows that these rules can shift. A 2026 club post names a 22-man squad at 4pm Tuesday and publishes a structured squad list including “Reserves” beyond the interchange. ⁸ This variation matters operationally: a production system should not hard-code “21-man Tuesday squads” as an invariant, and should store squad size and role labels as observed, season-by-season. ⁹

“Lineup effect” is not just “player present”. Rugby league roles are asymmetric: the spine concentrates decision-making and ball touches, interchange timing changes matchups, and late positional switches can change a player’s contribution even if the player remains selected. Rugby league performance research has already operationalised position-specific modelling, including separate distributions and models by position group and explicit modelling of interchange roles. ¹⁰

Licensing constraints shape what “public lineup data” means. League and club websites operate under terms that restrict use to personal, non-commercial contexts and limit reproduction without consent, which creates compliance risk for persistent scraping or redistribution. ¹¹ In parallel, commercial data providers exist: a 2018 press release states that STATS (now part of Stats Perform) signed a multi-year agreement as the official data provider and distributes live and post-match data via data feeds to partners. ¹² Rugby league research using event-level or detailed performance feeds frequently notes license and ethics constraints that prevent sharing raw data, which is a practical indicator that “full-fidelity” datasets are typically commercial rather than open. ¹³

Data sources and acquisition strategy

Primary sources that cover 2012–present lineups exist, but each source is strongest for a different “truth”: named squads, final 17s, positional labels, and last-minute changes.

A practical design is to store multiple lineup “versions” per match (e.g., named squad, 24-hour squad, final 17, actual minute participation where available), then let modelling choose which version to condition on depending on inference time. The monitoring layer should follow the publishing cadence and treat each fetch as an append-only event. ¹⁴

Candidate data sources comparison

Source (priority order)	What it covers well	Typical fields you can extract	Access pattern	Strengths	Key risks / caveats
Official league “team lists” news posts (2012– present archives)	Named squads and historical named lists; often includes “late mail” updates (at least in earlier years)	Player names, listed positions, interchange/ reserves labels, match metadata	HTML fetch + parser	Primary/ official; long historical depth visible in archives ¹⁵	Terms of use constrain reuse; page templates evolve across eras ¹⁶

Source (priority order)	What it covers well	Typical fields you can extract	Access pattern	Strengths	Key risks / caveats
Official club “team list” posts	Structured lineup lists; timestamps; often clearer separation of backs/ forwards/ interchange/ reserves	Player names, jersey numbers, position labels, squad size, timestamp	HTML fetch + parser	Structured lists with explicit positions and reserves; close to source of selection decisions ⁸	Template changes and occasional editorial text must be filtered; squad sizes can differ by event/ season ⁸
Official draw / match centre pages	Match metadata; links to related team lists; sometimes “Team Lists” print sections	Match IDs, venue/time, team names, related content	HTML fetch; sometimes requires backing API calls	Stable match identifiers and scheduling context ¹⁷	Some content is dynamically loaded; scraping may require reverse- engineering backing endpoints ¹⁸
Official stats pages	Aggregated player/team stats, often by competition and round	Player totals, team totals	Web app interface	Primary source of published stats exists ¹⁹	Often JS- driven; may fail without app context; not designed as a bulk export ¹⁹
Commercial data feeds (Stats Perform / Opta)	High-fidelity match, event, and performance data; potentially stints and substitutions	Events, minutes, substitutions, advanced metrics	Licensed API/ feed	Official- provider relationship and feed distribution to partners is documented ²⁰	Requires contract; data sharing often restricted by license and ethics ¹³

Source (priority order)	What it covers well	Typical fields you can extract	Access pattern	Strengths	Key risks / caveats
"Champion Data match centre API" as referenced by community tooling	Programmatic retrieval of match-centre style stats	Player/team stats; competition IDs	Unofficial programmatic access via wrappers	Public tooling documents that a match-centre API exists and can be called ²¹	Access may be brittle or unauthorised; credentials and terms may apply ²²
Community scraping packages (e.g., nrlR)	Accelerates discovery of endpoints and parsing patterns	Fixtures, some lineup scraping, stats wrappers	R package functions	Documents rate limiting, caching, and ethics guidance; includes lineup scraping entrypoints ²³	Not affiliated; can break when sites change; still bound by underlying site terms ²⁴
Media aggregators and analysis sites (Fox Sports, The Roar)	Human-readable summaries; injury and "team tips" content	Predicted lineups, commentary	HTML	Useful redundancy for changes and rumours (especially late mail style content) ²⁵	Not primary; paywalls and editorial bias; may not preserve historical structured data ²⁶

Near-term capture for "Thursday inference"

A robust design treats Thursday as "latest reliable snapshot before many weekend matches", not as a universal squad release time. In the 2017 description, the cut to 19 happens 24 hours pre-game, which means Wednesday for Thursday matches, Thursday for Friday matches, and so on. ⁶ The 2019 league post makes the same "24 hours before each game" statement. ⁷

Operationally, a weekly job should:

1. Pull the nominated squads as soon as they publish (historically Tuesday; in at least one 2026 club post this is timestamped 4pm). ²⁷
2. Pull the 24-hour updates per match in a match-aware schedule keyed off fixture times. ²⁸
3. Optionally pull the final 17 one hour pre-kick-off when latency allows, producing "late model refresh" predictions. ²⁸

Literature review on lineup and player-impact modelling

Rugby league has less public research on lineup-aware win probability than sports with rich public play-by-play, but several strands transfer directly.

Rugby league research has already built player contribution metrics tied to team success from multi-season professional data. One open-access study in *Research Quarterly for Exercise and Sport* (men's rugby league, 2018–2022 seasons) uses multiple machine learning algorithms and constructs “Wins Created”, “Losses Created”, and “Net Wins Added”; it reports strong match-winner classification performance and validates against awards and representative selection. ²⁹ A doctoral thesis in Australia reports a broader programme: literature review, player metrics in men's and women's rugby league, a match outcome model that incorporates lineup composition with team attributes (including an Elo component), and a decision support system that optimises lineups under constraints like availability and positional eligibility. ³⁰

Expected-points and expected-possession-value (EPV) modelling in rugby league provides a mechanism to create denser rewards than match outcome alone. Work using 2010–2013 NRL play-by-play estimated expected point outcomes by field location and possession context, highlighting how possession origin and field position change try likelihood and expected points. ³¹ Later work in rugby league uses Markov chain framing of sets and produces EPV zone models, explicitly discussing data licensing restrictions when using Stats Perform style feeds. ³² Bayesian EPV variants model smooth pitch surfaces and use hierarchical priors to share information between league and team levels, which mirrors the same partial pooling logic needed for player effects. ³³

In other invasion sports, lineup and player impact models largely fall into four families:

Adjusted plus-minus (APM) and its regularised variants estimate player contributions by regressing stint outcomes on who was on the field, addressing teammate and opponent confounding; regularisation is central to stability and out-of-sample behaviour. ³⁴ Soccer plus-minus work adapts the concept to low-scoring settings and introduces expected-goals-based plus-minus to reduce goal noise. ³⁵ Rating-system extensions (Elo/Glicko) add uncertainty and dynamic updates; Glicko explicitly introduces rating deviation as a reliability measure beyond Elo. ³⁶ Bayesian hierarchical models for match outcomes are widely used in football result prediction and provide a principled route to partial pooling across teams, seasons, and competitions. ³⁷ Representation learning (embeddings, graph models) encodes player interactions and co-occurrence in compact vectors; recent work explicitly describes swapping player embeddings to analyse how personnel changes affect match outcome predictions. ³⁸

Causal inference has become more explicit in sports analytics, including player evaluation. A causality-inspired APM model defines player value as a counterfactual score change under substitution, aligning with the core question “what is the effect of player X being in the team?”. ³⁹ Broader causal sports work emphasises that different estimands (ATE vs ATT) and matching/weighting designs change interpretation, which is directly relevant when player selection is non-random. ⁴⁰

Methods to infer lineup impact in an NRL win-probability model

A lineup-aware match model usually decomposes match outcome probability into three terms:

- team baseline strength (slow-moving, match-history learnable),
- lineup adjustment (fast-moving, player and position dependent), and
- uncertainty over the realised lineup at prediction time. ⁴¹

Modelling options and trade-offs

Approach	Minimum additional data	What it can learn	Interpretability	Main failure modes in NRL	Best role in your system
Additive player-presence model (regularised logistic / Poisson for points)	Final 17 and positions per match	“Player X in team” effects, position-weighted	High	Collinearity (teammates play together), small samples for fringe players	First upgrade; strong baseline for lineup effects ⁴²
Bayesian hierarchical player effects	Same as above, plus season/time structure	Partial pooling, time-varying effects, team switches	High to medium	Prior sensitivity; requires careful identifiability constraints	Best balance of rigour, stability, and uncertainty estimates ⁴³
APM / RAPM stint models	Substitution times and on-field stints	Context-adjusted on-field impact (per-minute)	Medium	Data access: stints often proprietary; sparse interchange exposure for some players	Optimal if you can license event/stint data; otherwise hard ⁴⁴
Elo/Glicko with lineup components	Team rating history + lineup features	Smooth dynamic updates; uncertainty via rating deviation	Medium	Hard to separate team vs player without shrinkage; rule changes across eras	Useful as a backbone for team baseline and uncertainty ⁴⁵

Approach	Minimum additional data	What it can learn	Interpretability	Main failure modes in NRL	Best role in your system
Embeddings (player/ position/ team) + shallow head	Same as additive, optionally event stats	Non-linear interactions; similarity; “swap player” counterfactuals	Medium to low	Overfitting in small leagues; harder diagnostics	Good second-phase once data pipeline is stable ³⁸
Causal APM / propensity weighting designs	Same as additive plus confounders	Closer to causal “effect of inclusion” interpretation	Medium	Unmeasured confounding (injury severity, coaching intent)	Use for sensitivity analysis of star-player effects ⁴⁶

Handling uncertainty: named squads vs the final 17

A match prediction made on Thursday should rarely assume the named squad equals the final 17, because selection rules and update windows explicitly permit changes up to one hour pre-kick-off. ²⁸ Your model can represent this uncertainty explicitly:

- Fit a selection model: $p(\text{plays} \mid \text{named, role, injury signals, travel, team context})$.
- Fit an outcome model conditioned on a realised final 17.
- Marginalise: $p(\text{win}) = \sum_{\ell} p(\text{win} \mid \ell) p(\ell \mid \text{current info})$, approximated via Monte Carlo over plausible final lineups. ⁴⁷

The 2017 description of selection stages gives direct structure for this two-stage approach (21/19/17 with limited exceptions). ⁶ The 2026 club example shows that even the named squad size can differ (22-man), so the selection model should condition on observed squad size rather than assuming it. ⁸

Matchup and interaction features that are specific to rugby league

Rugby league player contribution is context dependent. The men’s rugby league metric study explicitly splits variables by position and treats offence and defence separately, indicating that position-conditioned modelling is not optional if you want stable estimates. ⁴⁸ Practical lineup features that align with this include:

- position-weighted player ratings (spine vs middles vs edges vs outside backs),
- “spine continuity” features (same 1/6/7/9 combination as last match, last month),
- bench utility composition (e.g., presence of a hooker on bench) and interchange depth,
- team-level strength-of-schedule and opponent-strength adjustments when estimating player form.

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Expected points / EPV models provide a principled way to turn play and field position into a denser signal than win/loss, which can stabilise player and lineup estimates when match counts per season are modest.

50

Reinforcement learning for lineup optimisation and sequential decisions

Two RL problem shapes matter in rugby league: pre-match selection and in-match interchange.

Pre-match selection is closer to a *contextual bandit*: you observe a match context (opponent, venue, travel, injuries, recent form), pick a lineup action, then observe a reward (points differential, win probability change, or match result). The doctoral thesis described a decision support system that chooses optimal lineups under constraints such as availability and positional eligibility, which corresponds to a constrained decision problem even if implemented without RL.

30

In-match interchange is a sequential decision problem. The true physiological state (fatigue, knocks) is partially observed, so a POMDP is a closer formalisation than a fully observed MDP. Standard RL references define MDPs and POMDPs and show how belief states act as sufficient statistics under partial observability.

51

RL formulation and reward design

A useful modelling split is:

- **State**: time, score differential, current on-field lineup, interchange remaining, recent possession states, and latent fatigue proxy.
- **Action**: substitution decisions (who off, who on), or pre-match roster selection.
- **Reward**: dense rewards derived from expected points / EPV plus terminal match outcome, to reduce sparsity. Rugby league expected-points and EPV work shows that possession context and field position map to expected point equity, which naturally supports reward shaping.

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Offline RL is the default regime because you cannot safely or ethically explore random interchange policies in the real competition. Offline RL surveys and tutorials describe the “distribution shift” problem: learned policies can choose actions not supported by the dataset, causing extrapolation error, so conservative learning and safe policy improvement matter. Off-policy evaluation methods like doubly robust estimators are central when you want to evaluate a new policy using historical data logged under different behaviour policies.

53

Empirical work applying offline RL to professional basketball tactics indicates the feasibility of training policies from logged data, but that setting typically has richer possession-level data and more frequent decisions than rugby league, so transfer should be cautious.

54

flowchart TD

```
A[Context: opponent, venue, injuries, roster] --> B{Decision type}
B -->|Pre-match selection| C[Contextual bandit / constrained optimisation]
B -->|In-match interchange| D[POMDP: hidden fatigue + injury state]
```



```

C --> E[Action: choose squad/17/bench roles]
D --> F[Action: substitution or role change]
E --> G[Reward: points diff, win, EPV-shaped]
F --> G
G --> H[Offline RL / OPE needed for counterfactual evaluation]
H --> I[Policy constraints to stay within data support]

```

Simulation vs model-based planning

Planning methods need a transition model. In sports, “world models” often approximate how actions change win probability or expected outcome rather than simulate every micro-event. Action valuation work in football frames possessions with value functions, including deep RL approaches that estimate action-values from event or tracking data. ⁵⁵ Rugby league EPV models already treat sets as Markovian sequences, which is compatible with building a dynamics model at the level of field position and possession outcomes, even without full tracking. ⁵⁶

Practical pipeline, validation, and implementation roadmap

The pipeline should separate three concerns: (i) ingestion and canonicalisation, (ii) feature construction at a chosen “as-of time”, and (iii) modelling and evaluation.

```

flowchart LR
    subgraph Sources
        S1[League team-list posts]
        S2[Club team-list posts]
        S3[Fixture/draw pages]
        S4[Commercial feed (optional)]
    end

    S1 --> I[Ingest + snapshot raw HTML/JSON]
    S2 --> I
    S3 --> I
    S4 --> I

    I --> P[Parse to canonical schema\nmatch_id, team_id, player_id,\nlisted_position, status, timestamp]
    P --> Q[Entity resolution\nname->player_id\nteam changes\nposition\nnormalisation]
    Q --> L[Lineup store\nversioned by timestamp\n(named/24h/final/actual)]
    L --> F[Feature store\n(as-of time)]
    F --> M[Model training\n(time-respecting splits)]
    F --> R[Realtime inference\nThursday snapshot + uncertainty]
    M --> E[Evaluation\nlog loss, Brier, calibration,\nseason holdouts]
    R --> O[Outputs\nwin prob + cred intervals\nplayer contribution deltas]

```

Feature engineering blueprint

A compact but high-yield set of features aligns with what rugby league performance work already models: offence/defence separation and position grouping, plus adjustments for opponent strength. ²⁹

Feature block	Examples	Why it helps	Data dependency
Player availability and selection uncertainty	For each named player: probability of making final 17; bench vs reserve probabilities	Converts squad lists into quantified uncertainty aligned with NRL selection stages ²⁸	Historical named-vs-final mapping
Lineup strength aggregates	Sum/mean of player ratings by position group; spine-only strength; bench strength	Captures “who is playing” in a low-dimensional way with shrinkage ¹⁰	Player rating system, position labels
Matchup deltas	Home spine strength minus away spine strength; edge forward mismatch index	Encodes asymmetric game influence of key roles and matchups ²⁹	Position mapping + ratings
Continuity and cohesion	Same halves pairing as last week; games played together for spine; recent lineup stability	Approximates interaction effects with controlled dimensionality ⁵⁷	Historical lineups
Context beyond lineups	Rest days, travel proxies, venue; baseline team rating	Preserves what your current match-history model already captures ⁵⁸	Fixture + team history
EPV/expected-points derived (optional)	Expected points from start sets, field position; EPV-based attack/defence profiles	Provides denser signal for learning player impacts when outcomes are noisy ⁵⁰	Play-by-play or event feed

Training, validation, and evaluation metrics

A time-ordered evaluation is essential because team rosters, rules, and playing styles shift across 2012–present. Rugby league research already reports order-preserved testing as a distinct evaluation mode for match prediction models. ³⁰

A defensible protocol:

- **Primary split:** rolling-origin evaluation (train through season t , test on season $t+1$), repeated for multiple years. ³⁰
- **Secondary split:** within-season forecasting (train up to round r , test on rounds $r+1\dots$), matching “Thursday inference” operational reality. ²⁸

- **Metrics:** log loss and Brier score for probabilistic accuracy; reliability diagrams / calibration error; and discrimination metrics (AUC) as secondary. These are standard for probabilistic forecasts and align with the “win probability” output target. ⁵⁹

Missing data, late changes, and player movement

Late changes should not be treated as noise. The official selection cadence explicitly allows changes near kick-off and, in limited cases, outside the named squad for approved reasons. ²⁸ Store every observed update as a timestamped lineup version, and train models to condition on “as-of time” because the information set differs materially between Tuesday, Thursday, and match-day. ⁶⁰

Transfers and position changes create identifiability problems if you model “player value” as static. Partial pooling helps: hierarchical priors shrink uncertain player effects toward position and team baselines, and time-varying components can capture form changes. This is consistent with how Bayesian hierarchical sports models handle heterogeneity and how causal APM frames substitution effects under uncertainty. ⁴³

Recommended experiments and ablations

A small set of ablations can isolate what lineup information adds:

- Baseline match-history model vs match-history + “named squad only” features at the same cut-off time. ⁶¹
- Replace binary “player present” with position-weighted player ratings and compare calibration and log loss. ⁶²
- Deterministic final-17 assumption vs probabilistic marginalisation over possible final 17s (Monte Carlo). ⁶³
- Additivity-only vs additivity + limited interaction terms (spine pairings), with strong regularisation, to test whether synergy is learnable without overfit. ⁶⁴
- Sensitivity analysis: causal-inspired adjustment (weighting/matching) for a small set of star-player treatments to test confounding magnitude. ⁴⁶

Prioritised implementation roadmap with effort and risks

Phase	Deliverable	Estimated effort (no stack constraints)	Main risks	Risk controls
Foundation	Canonical lineup store (2012–present) from official league + club posts; versioned snapshots	1–3 weeks	Template drift; name collisions	Snapshot raw pages; regression tests; deterministic name→ID rules with manual override queue ⁶⁵

Phase	Deliverable	Estimated effort (no stack constraints)	Main risks	Risk controls
First modelling lift	Regularised additive player/position model; Thursday inference using named squads	2–4 weeks	Overfitting; collinearity	Ridge/Lasso + position pooling; strict time splits; ablation gating ⁶⁶
Uncertainty upgrade	Selection-probability model + marginalised win probabilities; credible intervals	3–6 weeks	Mis-calibration; sparse historical examples	Calibration curves; hierarchical priors; backtesting by “as-of time” ⁶⁷
High-fidelity impacts (optional)	Stint-based APM/ RAPM from substitution timelines	6–12+ weeks	Data access; licensing	Consider licensing commercial feeds; otherwise deprioritise ⁶⁸
RL exploration (research track)	Contextual bandit for selection; POMDP for interchange with OPE	8–16+ weeks	Dataset support mismatch; unreliable OPE	Conservative offline RL; doubly robust OPE; safe policy improvement constraints ⁶⁹

Deployment considerations: latency, compute, and monitoring

The dataset scale is modest: 2012–present yields only a few thousand matches, so model training is not compute-bound unless you add heavy neural components. The operational risk is data freshness and schema breakage, not GPU time. Community tooling explicitly emphasises rate limiting, caching, and handling missing elements as first-class engineering issues, which matches the realities of lineup scraping. ⁷⁰

“Thursday inference” latency is dominated by ingestion (waiting for squad updates) rather than feature computation. A production system benefits from two inference modes: (i) scheduled refresh keyed to the 24-hour update windows, and (ii) rapid re-score when last-minute lineup changes publish. The selection-stage descriptions justify this design. ⁷¹

Compliance is a non-technical blocker. The league site’s terms include non-commercial use language and restrictions on reproducing site material without consent, which is incompatible with broad redistribution of scraped lineups. If predictions are used beyond personal research, licensing discussions with official data providers are the cleanest route. ⁷²

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